

SHANMUGARAJ

Full-Stack Developer (MERN) | Software Application Development

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SUMMARY

High-performance Software Development Engineer proficient in JavaScript, TypeScript, C++, and Python. Experienced in designing scalable backend architectures, custom encoding algorithms, and high-frequency caching layers. Strong grasp of fundamental computer science concepts including OOP, DBMS, OS, and the Software Development Lifecycle (SDLC) focused on writing clean, modular, and maintainable code.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, C++, SQL, Python, Java

Frontend: React.js, Next.js, HTML5, CSS3

Backend & System Design: Node.js, Express.js, RESTful APIs, Microservices, MVC Architecture, Caching (Redis)

Databases: PostgreSQL, MongoDB, Query Optimization, Relational Schema Design

Developer Tools: Git, GitHub, Postman, Vercel, Docker/CI-CD

EDUCATION

B.E., Electronics & Communication Engineering

2021 - 2025 (Graduated)

Jerusalem College of Engineering, Chennai

CGPA: 7.5/10 | No Arrears | Coursework: DSA, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Software Intern | *NSIC - Technical Services Center*

2024

- Developed and debugged C/C++ applications for microcontroller platforms, building strong foundations in low-level programming, memory management, and systematic root-cause analysis.
- Integrated AI/ML inference into embedded systems and performed structured testing and validation, applying the same debugging discipline to full-stack project development.

PROJECTS

Tiny-URL Scalable URL Shortener | *Authorization Node.js · Express.js · MongoDB · Redis · REST API · MVC · Caching*

[GitHub](#)

- Built a core URL shortening service with Node.js and Express, generating unique short codes and handling high-volume redirects through RESTful endpoints.
- Integrated Redis caching to serve frequently accessed redirects without hitting the database, cutting average lookup latency.
- Added URL authorization to secure link creation and restrict access to owned links.

Task Flow - Team Task Manager | *React.js, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, RBAC*

[Live Demo](#) | [GitHub](#)

- Built and deployed a full-stack task management platform with a Kanban board, task filtering, overdue task tracking, and role-based collaboration for team project management.
- Designed MVC RESTful APIs with JWT authentication and admin/member RBAC; optimized PostgreSQL queries using Prisma ORM to improve performance and scalability.
- Kept the codebase clean and production-scalable using modular React components and Git feature branching.

College Transport Tracking System | *Flutter, Dart, Google Maps API, REST API, Real-Time Geolocation*

- Independently learned Flutter/Dart to build a cross-platform live vehicle-tracking app using the Google Maps API; diagnosed and fixed real-time map lag by optimizing REST API polling frequency.

PROBLEM SOLVING

- 150+ LeetCode problems solved with deep focus on time/space complexity analysis applied to backend optimization.
- Studying distributed systems and system design fundamentals: scalability, caching, load balancing, database sharding, microservices.

ACHIEVEMENTS

- 2nd Place, College Web Development Competition - built a fully responsive, functional web app recognized among all competing teams.
- Deployed 2 live full-stack applications on Vercel featuring RBAC, JWT authentication, and both SQL and NoSQL databases.